



GURU TEGH BAHADUR PUBLIC SCHOOL, DURGAPUR
UNIT TEST – II : SESSION 2026-27

CLASS : XI

MAXIMUM MARKS : 50

SUBJECT : COMPUTER SCIENCE (083) TIME ALLOWED : 1 Hr. 45 Min.

General Instructions:

1. This question paper contains five sections, Section A to E.
2. All questions are compulsory.
3. All programming questions are to be answered using Python Language only.
4. Section A has no multiple-choice questions. Write only the missing word, the answer or the exact output. Write all the sub-parts of a question together.

SECTION - A		
Q.1	Answer the following:	(1 × 13 = 13)
(a)	Name the logic gate whose output is 1 only when its two inputs are different.	1
(b)	The Boolean expression $A + A'B$ simplifies to _____.	1
(c)	The output of a NAND gate is 0 only when _____.	1
(d)	Write the output of the following: <pre>print("Welcome To My Blog"[2:6] + "Welcome To My Blog"[5:9])</pre>	1
(e)	Write the output of the following: <pre>message = "World Peace" print(message[-2::-2])</pre>	1
(f)	Write the output of the following: <pre>text = "PYTHONPROGRAM" text = text.replace('PY', '#') print(text)</pre>	1
(g)	Write the output of the following: <pre>country = 'International' print(country.split("\n"))</pre>	1
(h)	Write the output of the following: <pre>L = ['TIC', 'TAC'] print(L[::-1])</pre>	1

(i)	The _____ method of a list removes the first occurrence of a given value from the list.	1
(j)	What will be the result of executing the following code? <pre>game = "Olympic2024" print(game.index("C"))</pre>	1
(k)	Name the Python statement that does nothing and is used as a placeholder where a statement is syntactically required.	1
(l)	How many times will "Hi" be printed by the following code? <pre>for i in range(2, 20, 4): print("Hi")</pre>	1
(m)	Write the output of the following: <pre>print(19 < 11 and 29 > 19 or not 75 > 30)</pre>	1
SECTION - B		
Q.2	Prove De Morgan's first law, $(A + B)' = A' \cdot B'$, using a truth table.	(2)
Q.3	Answer using Python built-in methods/functions only: (a) Write a statement to find the index of the first occurrence of the substring "good" in a string named review . (b) Write a statement to sort the elements of list L1 in descending order.	(1+1)
Q.4	Write the output of the following code: <pre>L = [10, 20, 30, 40, 50] L.insert(2, 25) L.pop() L.extend([5, 15]) print(L) print(L.index(30), L.count(20))</pre>	(2)
SECTION - C		
Q.5	For the XOR gate with inputs A and B: (a) Write its Boolean expression using only AND, OR and NOT operations. (b) Draw its truth table. (c) Draw its logic circuit using only AND, OR and NOT gates.	(1+1+1)

Q.6	Predict the output of the following Python code: <pre> s1 = "SQP-25" s2 = "" i = 0 while i < len(s1): if s1[i] >= '0' and s1[i] <= '9': Num = int(s1[i]) Num -= 1 s2 = s2 + str(Num) elif s1[i] >= 'A' and s1[i] <= 'Z': s2 = s2 + s1[i+1] else: s2 = s2 + '^' i += 1 print(s2) </pre>	(3)
Q.7	Write a program to input n numbers into a list and display the largest number, the smallest number and the mean of the numbers, <u>without</u> using the built-in functions max(), min() and sum().	(3)
SECTION - D		
Q.8	(a) Simplify the following Boolean expression using Boolean laws (show each step): $(A + B) \cdot (A + C)$ (b) Verify the following using a truth table: $A + A'B = A + B$	(2+2)
Q.9	Write a program that inputs a string from the user and: (a) checks whether the string is a palindrome or not (ignoring the case of letters), and (b) counts and displays the number of words in the string that begin with a vowel.	(2+2)
Q.10	Write the output of the following code segments: (a) <pre> a = "Year 2022 at All the best" a = a.split('2') b = a[0] + ". " + a[1] + ". " + a[3] print(b) </pre> (b) <pre> pride = "#G20 Presidency" print(pride[-2:2:-2]) </pre> (c) <pre> st = 'Waterskiing is thrilling!' print(st.split("i")) </pre> (d) <pre> L = ["India", "Incredible", "Bharat"] print(L[1][0] + L[2][-1]) </pre>	(1×4)

SECTION - E		
Q.11	Write a program that inputs a list of n integers from the user and then: (a) displays the list after removing the duplicate elements, keeping only the first occurrence of each element; (b) asks the user for a number and performs a linear search for it in the original list, displaying its position if found, otherwise displaying "Not found".	(2+3)
Q.12	(a) Predict the output of the following Python code: <pre>wildlife_sanctuary = ["Kaziranga", "Ranthambhore", "Jim Corbett", "Sundarbans", "Periyar", "Gir", "Bandipur"] output = [] for sanctuary in wildlife_sanctuary: if sanctuary[-1] in 'aeiou': output.append(sanctuary[0].upper()) print(output)</pre> (b) Write a program using a while loop to input a positive integer and display the sum of its digits and its reverse. For example, for 1234 the output should be: Sum = 10, Reverse = 4321	(2+3)

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